Abdurro'uf

Curriculum Vitae

Education

2015–2018 **Ph.D.,** Astronomical Institute, Tohoku University, Japan

Adviser: Prof. Masayuki Akiyama

Thesis title: *Spatially resolved stellar mass buildup and quenching of star formation in massive disk galaxies over the last 10 Gyrs*

2013–2015 **M.Sc.,** Astronomical Institute, Tohoku University, Japan

Adviser: Prof. Masayuki Akiyama

Thesis title: *Star formation mode of galaxies in the local universe with spatially-resolved star formation rate and stellar mass*

2009–2013 **B.Sc.,** Physics Department, Brawijaya University, Indonesia

Work experience

2018–present Postdoctoral researcher at the Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taiwan

Research interests

- Galaxy formation and evolution, in particular, evolution of the galaxy structures and mechanism of the galaxy quenching
- Modeling of Galaxy spectral energy distribution (SED) and development of statistical method to extract properties of galaxies from their SEDs
- Simulation of galaxy formation and evolution
- Observational Cosmology

Awards

2013–2018 **Japanese Government (Ministry of Education, Culture, Sports, Science, and Technology, MEXT) Scholarship** for Master and Doctor studies

2013 **Silver medal** in physics field on the National Olympiad of Mathematics and Natural Sciences for undergraduate student (ON MIPA-PT), Yogyakarta, Indonesia

Level of recognition: National

No. 137, Sec. 1, Zhongzheng East Road
Tamsui District, New Taipei City, Taiwan
(+886)905606672 • *abdurrouf@asiaa.sinica.edu.tw*
https://aabdurrouf.github.io/ • *aabdurrouf*
Postdoctoral researcher at ASIAA

2012 **Silver medal** in physics field on the National Olympiad of Mathematics and Natural Sciences for undergraduate student (ON MIPA-PT), Surabaya, Indonesia

Level of recognition: National

Languages

Indonesian/Bahasa (native), Javanese ethnic language (native), English (fluent)

Publication

Full list of my publication in ADS, Google Scholar

Refereed Journal

2021 "*Introducing piXedfit: A Spectral Energy Distribution Fitting Code Designed for Resolved Sources*", (**Abdurro'uf**), Yen-Ting Lin, Po-Feng Wu, and Masayuki Akiyama, The Astrophysical Journal Supplement Series (ApJS), Volume 254, Issue 1, id.15, 32 pp., DOI:10.3847/1538-4365/abebe2

Publisher, ADS, arXiv

2021 "*Extremely Low Molecular Gas Content in the Vicinity of a Red Nugget Galaxy at* $z = 1.91$", T. Morishita, Q. D'Amato, L. E. Abramson, **Abdurro'uf**, M. Stiavelli, R. A. Lucas , The Astrophysical Journal (ApJ), Volume 908, Issue 2, id.163, 9 pp., DOI:10.3847/1538-4357/abd800

Publisher, ADS, arXiv

2020 "*Tracing the Coevolution Path of Supermassive Black Holes and Spheroids with AKARI-selected Ultraluminous IR Galaxies at Intermediate Redshifts*", Xiaoyang Chen, Masayuki Akiyama, Kohei Ichikawa, Hirofumi Noda, Yoshiki Toba, Issei Yamamura, Toshihiro Kawaguchi, **Abdurro'uf**, Mitsuru Kokubo,The Astrophysical Journal (ApJ), Volume 900, Issue 1, id.51, 28 pp., DOI:10.3847/1538-4357/aba599

Publisher, ADS, arXiv

2019 "*Discovery of a strong ionized-gas outflow in an AKARI-selected ultra-luminous infrared galaxy at* $z = 0.5$", Xiaoyang Chen, Masayuki Akiyama, Hirofumi Noda, **Abdurro'uf**, Yoshiki Toba, Issei Yamamura, Toshihiro Kawaguchi, Mitsuru Kokubo, Kohei Ichikawa, Publications of the Astronomical Society of Japan (PASJ), Volume 71, Issue 2, id.29, DOI:10.1093/pasj/psz002

Publisher, ADS, arXiv

2018 "*Evolution of spatially resolved star formation main sequence and surface density profiles in massive disc galaxies at* $0 \lesssim z \lesssim 1$*: inside-out stellar mass buildup and quenching*", (**Abdurro'uf**) & Masayuki Akiyama, Monthly Notices of the Royal Astronomical Society (MNRAS), Volume 479, Issue 4, p.5083-5100, DOI:10.1093/mnras/sty1771

Publisher, ADS, arXiv

2017 "*Understanding the scatter in the spatially resolved star formation main sequence of local massive spiral galaxies*", (**Abdurro'uf**) & Masayuki Akiyama, Monthly Notices of the Royal Astronomical Society (MNRAS), Volume 469, Issue 3, p.2806-2820, DOI:10.1093/mnras/stx936

Publisher, ADS, arXiv

Non-refereed Journal and Proceeding

2017 "*Spatially resolved stellar mass buildup and quenching in massive disk galaxies over the last 10 Gyr revealed with spatially resolved SED fitting*", (**Abdurro'uf**) & Masayuki Akiyama, Challenges in Panchromatic Modelling with Next Generation Facilities. Proceedings of the International Astronomical Union, Volume 341, pp. 55-59, DOI:10.1017/S1743921319002680

Publisher, ADS, arXiv

Book

2014 "*Penyelesaian Soal OSN Pertamina Bidang Fisika*", Abdurrouf, M Shoufie Ukhtary, **Abdurro'uf**, Universitas Brawijaya Press

Website

Presentation

Oral presentation

2021 (**Invited**) "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", **National Central University, Graduate Institute of Astronomy Colloquium**, NCU, Taiwan: April, 2021

Website

2021 (**Invited**) "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", **CANDELS SED fitting working group teleconference**, virtual (USA-Canada-Taiwan): March, 2021

Website, CANDELS

2021 (**Invited**) "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", **Insitut Teknologi Bandung (ITB; Bandung Institute of Technology) Astronomy Department Colloquium**, ITB, Indonesia (virtual): February, 2021

Website, Video

2021 "*Spatially resolved spectrophotometric SED fitting of galaxies with piXedfit*", **Japan-Korea-Taiwan Joint Galaxy Evolution Workshop 2020**, ASIAA, Taiwan (virtual): February, 2021

Website

2020 (**Invited**) "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", **National Tsing Hua University (NTHU), Institute of Astronomy Colloquium**, Taiwan: June, 2020

Website

2019 "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", **Galaxy Formation and Evolution Across Cosmic Time**, ASIAA, Taiwan: December, 2019

Website

2019 "*Study spatially resolved stellar population properties of galaxies with spatially resolved SED fitting*", **ASIAA colloquium**, ASIAA, Taiwan: June, 2019

Website

2018 "*Spatially resolved stellar mass buildup and quenching in massive disk gaalxies over the last 10 Gyr revealed with spatially resolved SED fitting*", **IAU Symposium 341: PanModel2018: Challenges in panchromatic galaxy modelling with next generation facilities**, Osaka University, Japan: November, 2018

Website, Proceeding

2018 "*Spatially resolved stellar mass buildup and quenching in massive disk galaxies at* $0 \lesssim z \lesssim 1$", **Galaxy Evolution Workshop**, Ehime University, Japan: June, 2018

Website, Proceeding

2018 "*Spatially resolved stellar mass buildup and quenching in massive disk galaxies at* $0 \lesssim z \lesssim 1$", **Astronomical Institute's colloquium**, Tohoku University, Japan: April, 2018

Website

2017 "*Evolution of the spatially resolved star formation main sequence of massive disk galaxies at* $0 \lesssim z \lesssim 1$ ", **Sharp views of galaxy formation and evolution, Japan-Germany workshop**, Tohoku University, Japan: November, 2017

Website

2017 (**Invited**) "*Understanding the scatter in the spatially resolved star formation main sequence*", **Galaxy workshop (Galshop)**, held in teleconference among NAOJ Mitaka-campus, Subaru telescope, and Tohoku University: June, 2017

Website, Proceeding

2016 "*Spatially resolved star formation rate and stellar mass of local massive spiral galaxies: understanding the scatter in spatially resolved star formation main sequence*", **How galaxies form stars conference**, Stockholm University, Sweden: August, 2016

Website, Proceeding

2016 "*Spatially-resolved star formation rate and stellar mass of galaxies in the local universe: quantifying the inside-out scenario of disk galaxies formation*", **The 3rd Galaxy evolution workshop**, Tohoku University, Japan: June, 2016

Website, Proceeding

2015 "*Spatially resolved star formation rate density and stellar mass density of galaxies in the local universe*", **Astronomical Institute's Colloquium**, Tohoku University, Japan: May, 2015

Website

Poster presentation

2020 "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", Abdurro'uf, Yen-Ting Lin, Po-Feng Wu, Masayuki Akiyama, **2020 Annual Meeting of The Astronomical Society of the Republic of China (Taiwan)**, ASIAA, Taiwan: September, 2020

Website

2019 "*Spatially resolving properties of galaxies across cosmic time with piXedfit*", Abdurro'uf, Yen-Ting Lin, Masayuki Akiyama, **The art of measuring galaxy physical properties**, INAF-IASF, Italy: November, 2019

Website, Proceeding

2018 "*A systematic search for ULIRGs/HyLIRGs at intermediate redshifts with optically-faint AKARI FIS Bright sources*", Xiaoyang Chen, Masayuki Akiyama, Hirofumi Noda, Tatsuro Watanabe, Abdurro'uf, Yoshiki Toba, and Issei Yamamura, **Subaru Users' Meeting FY2017**, NAOJ Miataka Campus, Japan: January, 2018

Website

2017 "*A systematic search for ULIRGs/HyLIRGs at intermediate redshifts with optically-faint AKARI FIS Bright sources*", Xiaoyang Chen, Masayuki Akiyama, Hirofumi Noda, Tatsuro Watanabe, Abdurro'uf, Yoshiki Toba, and Issei Yamamura, **East-Asia AGN Workshop 2017**, Kagoshima University, Kagoshima, Japan: December, 2017

Website

2017 "*A systematic search for ULIRGs/HyLIRGs at intermediate redshifts with optically-faint AKARI FIS Bright sources*", Xiaoyang Chen, Masayuki Akiyama, Hirofumi Noda, Tatsuro Watanabe, Abdurro'uf, Yoshiki Toba, and Issei Yamamura, **AKARI Conference 2017**, The University of Tokyo, Japan: October, 2017.

Website, Proceeding

2016 "*Spatially resolved star-formation main sequence in the local massive spiral galaxies*", Abdurro'uf and Masayuki Akiyama, **Panoramas of the evolving cosmos, The 6th Subaru international conference**, Hiroshima, Japan: November, 2016.

Website, Proceeding (P17)

2015 "*Spatially resolved star formation rate density and stellar mass density of massive galaxies in the local universe*", Abdurro'uf and Masayuki Akiyama, **Science and Life Science Joint Symposium 2015**, Tohoku University, Japan: February, 2015

Telescope time

2021 Co-Investigator, 8 nights observations at the SEIMEI telescope, Okayama, Japan: May, 2021

"*A new population of extreme starburst galaxies at intermediate redshifts*", Masayuki Akiyama, Xiaoyang Chen, Yoshiki Toba, Mitsuru Kokubo, **Abdurro'uf**, Yamamura Issei, Toshihiro Kawaguchi, Hirofumi Noda, Koki Terao, Kohei Ichikawa

2020 Co-Investigator, 8 nights observations at the SEIMEI telescope, Okayama, Japan: December, 2020

"*A new population of extreme starburst galaxies at intermediate redshifts*", Masayuki Akiyama, Xiaoyang Chen, Yoshiki Toba, Mitsuru Kokubo, **Abdurro'uf**, Yamamura Issei, Toshihiro Kawaguchi, Hirofumi Noda, Koki Terao, Kohei Ichikawa

2020 Co-Investigator, 3 nights observations at MMT telescope, Arizona, USA: October, 2020

"*Unvealing the nature of isolated massive slow-rotating early-type galaxies*", Hung-Jin Huang, Xiaohui Fan, **Abdurro'uf**, Yen-Ting Lin, Tim Eifler

2020 Co-Investigator, 4 nights observations at MMT telescope, Arizona, USA: June, 2020

"*Unvealing the nature of isolated massive slow-rotating early-type galaxies*", Hung-Jin Huang, Xiaohui Fan, **Abdurro'uf**, Yen-Ting Lin, Tim Eifler

2017 Co-Investigator, 1 night observation at the Subaru Telescope, Hilo, USA

"*New Search for Luminous Type-2 QSOs at Intermediate Redshifts*", Masayuki Akiyama, Hirofumi Noda, Tatsuro Watanabe, **Abdurro'uf**, Yongming Liang, Tian Qiu, Juming Yu, Guangwen Chen

Software experience

Programming languages: Python (proficient), C++ (proficient), C, Delphi

Experiences in developing and using existing softwares:

1. Developed **piXedfit**, a new spectral energy distribution (SED) fitting code designed for resolved galaxies
2. Developed **Accurate Hijri Calculator (AHC)**, a software for calculating a map of visibility of a new crescent at a sunset time that mark a begining of a month in the Islamic lunar calendar. I developed this software for my Bachelor thesis project
3. Experienced working with IRAF, SExtractor, emcee, Astropy, Photutils, FSPS

Attendance in summer school and workshop

2019 **11th Prime Focus Spectrograph (PFS) collaboration meeting 2019**, Caltech, Pasadena, USA: December, 2019

2017 **TIARA summer school on Astrostatistics and big data**, ASIAA, Taipei, Taiwan: September, 2017.

2016 **Thirty Meter Telescope (TMT) Science Forum**, Kyoto, Japan: May, 2016.

No. 137, Sec. 1, Zhongzheng East Road
Tamsui District, New Taipei City, Taiwan
(+886)905606672 • abdurrouf@asiaa.sinica.edu.tw
https://aabdurrouf.github.io/ • aabdurrouf
Postdoctoral researcher at ASIAA